

Team Exodus

PharmaGuard AI

IBM Bob AI Hackathon — Track: AI



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Pharmacovigilance &
Regulatory AI





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The Problem

Pharmacovigilance teams spend weeks manually correlating adverse-event signals with regulatory submission gaps. PharmaGuard AI unifies FDA FAERS signal detection and ICH M4 CTD readiness checking into a single, fully working prototype — making safety analysis faster, traceable, and submission-ready.

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Technical Architecture

A fully implemented, end-to-end prototype built on a modern, production-grade stack. Every layer — from data ingestion to regulatory output — is operational and tested, with 956 backend tests passing.

Python / FastAPI Backend

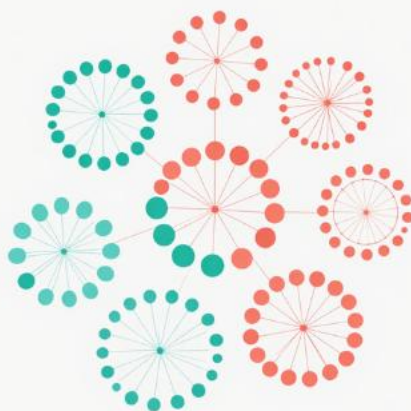
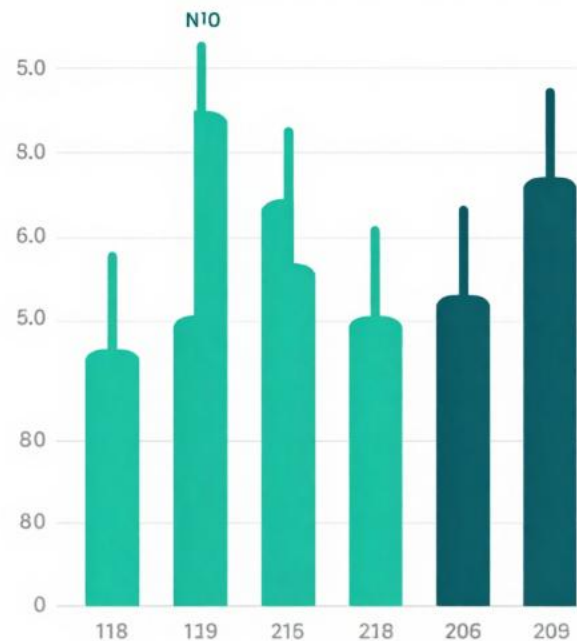
FDA FAERS Pipeline

React + TypeScript Frontend

IBM Bob MCP Server (6 Tools)



Adverse drug event signals



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Signal Detection

PRR-Based Analysis

FDA FAERS data pipeline drives PRR-based adverse-event detection, flagging statistically significant drug-event pairs. Temporal trend analysis tracks signal evolution over time, while multi-dimensional priority scoring and automated clustering surface the highest-risk signals for immediate regulatory review.



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Signal Details

From Evidence to Action

Evidence & Trend Context

Each signal surfaces the full adverse-event evidence trail: PRR score, case count, reporting timeline, and drug-event co-occurrence. Temporal trend analysis plots reporting frequency over time, revealing whether a signal is emerging, stable, or declining — giving analysts the longitudinal context needed for confident assessment.

Priority Scoring & Clustering

A multi-dimensional priority score weighs signal strength, case severity, and regulatory exposure. Signals are then clustered by drug class and event type, so analysts see related risks together — reducing redundant review and surfacing systemic safety patterns across the portfolio.

Decision-Support Outputs

Each Signal Details view generates a structured Action Brief with recommended next steps, CTD section traceability links, and regulatory gap flags. IBM watsonx.ai provides narrative summaries with deterministic fallback — ensuring consistent, audit-ready outputs every time.



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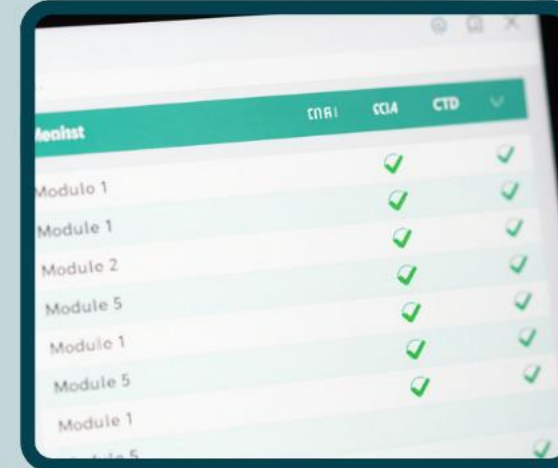
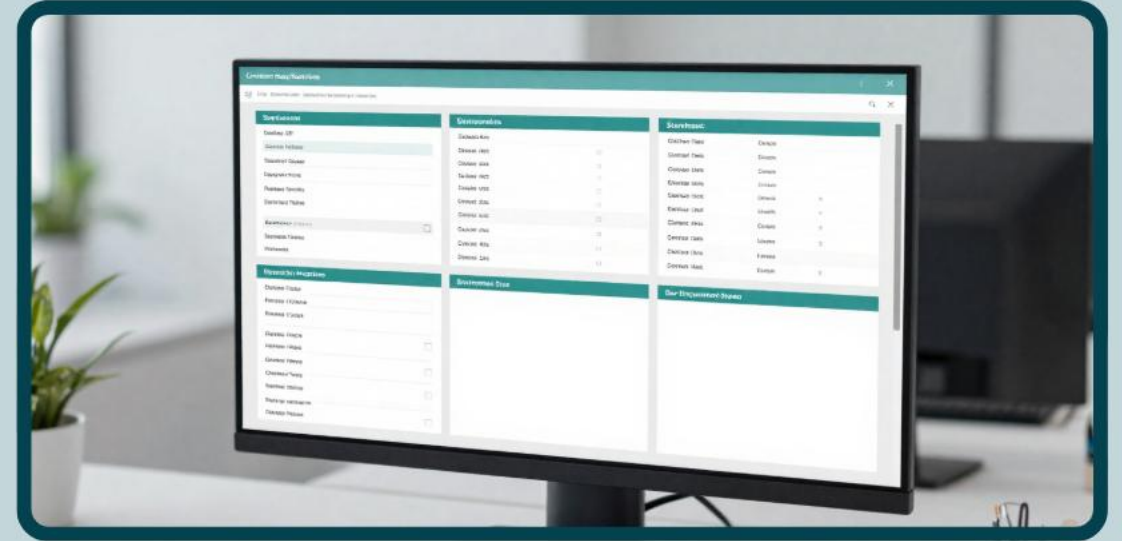
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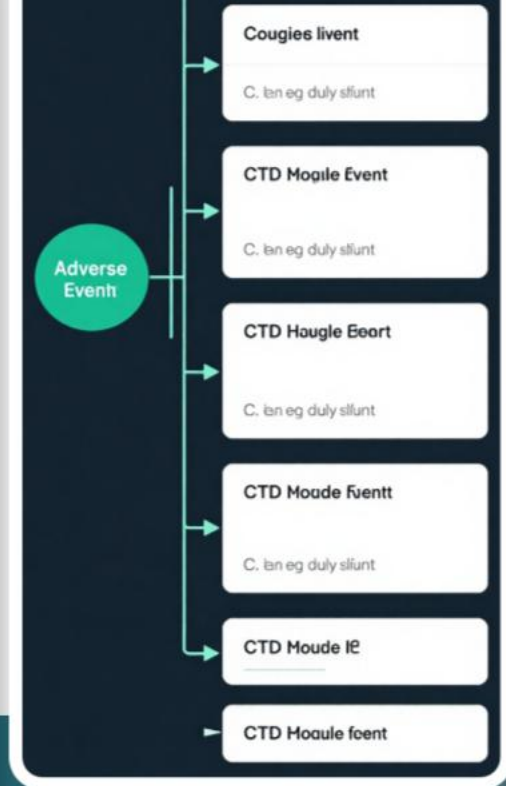
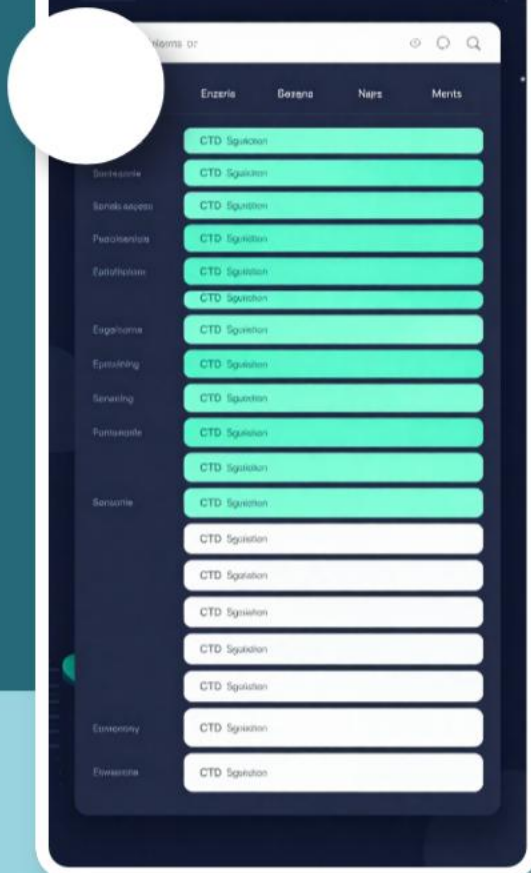
Submission Readiness

ICH M4 CTD Compliance Engine

PharmaGuard AI automates ICH M4 CTD submission readiness assessment by parsing regulatory dossiers and matching content against required sections. The engine evaluates completeness across Module 1–5, identifies missing or incomplete sections, and generates a structured gap report. Safety signals are automatically traced to their corresponding CTD sections, ensuring full traceability from detection to submission.



Automated CTD gap reporting with section-level traceability from safety signals to submission modules.

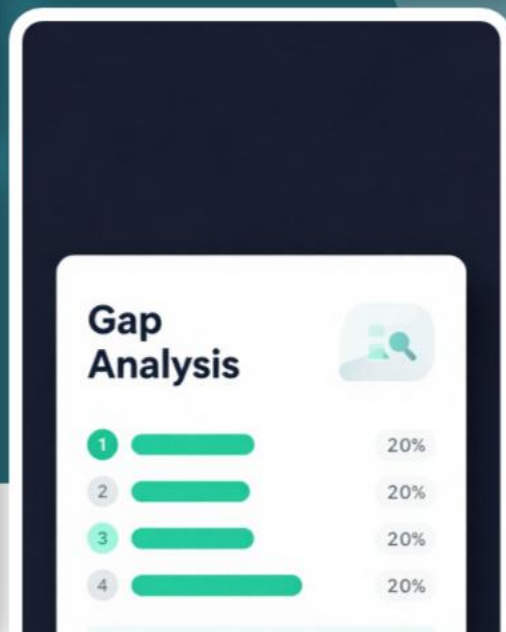
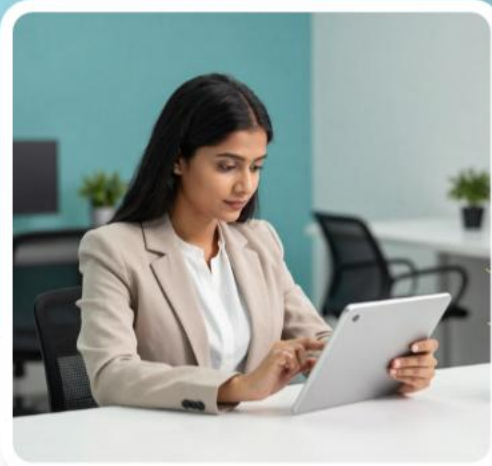


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Traceability & Gap Analysis

CTD Compliance Engine

Safety signals are automatically traced to ICH M4 CTD sections. Regulatory gap reports and Action Briefs surface missing documentation before submission.

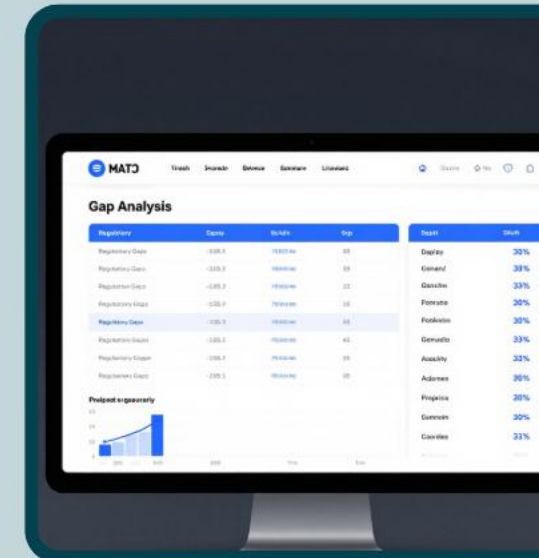
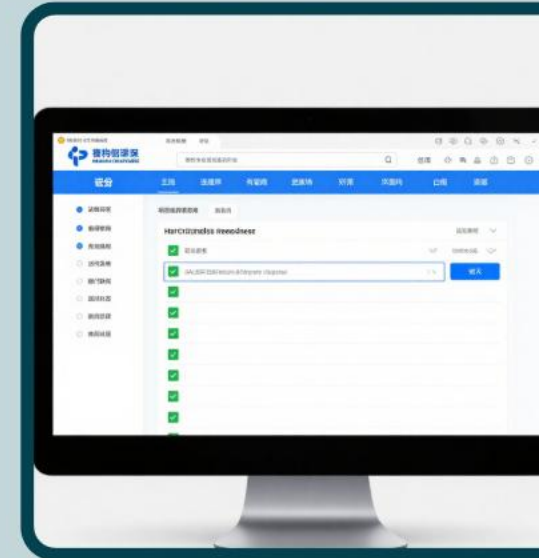


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Frontend Experience

Six Implemented Pages

Dashboard — live signal metrics & KPI overview. Signal Detection — PRR-based adverse event signals from FDA FAERS data. Signal Details — deep-dive temporal trends & multi-dimensional scoring. Submission Readiness — ICH M4 CTD compliance checker. Gap Report — regulatory gap identification & traceability. Action Brief — prioritized next-step recommendations.

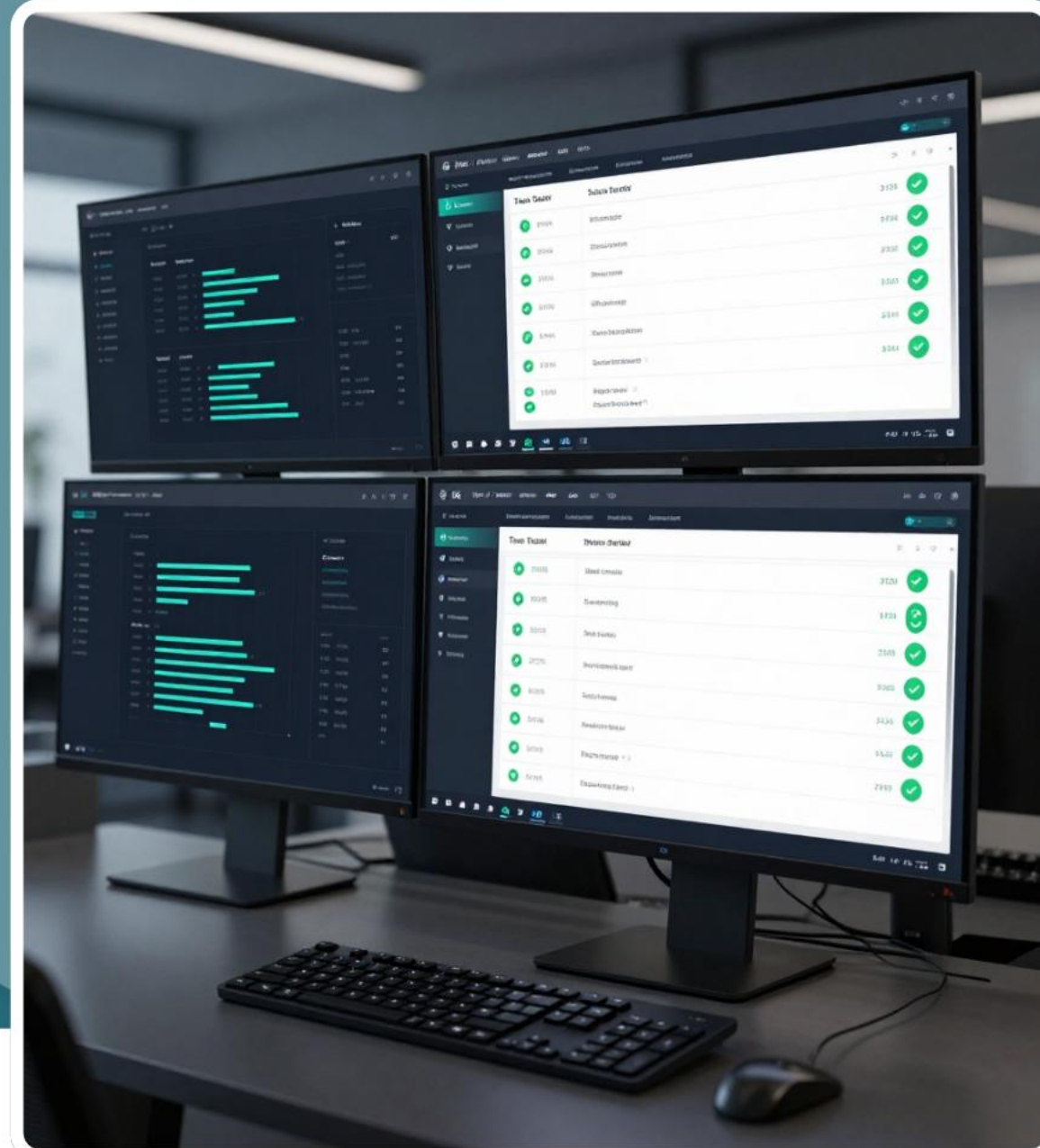


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Validation & AI Integration

Quality, Reliability & Intelligent Orchestration

956 backend tests passing with full coverage across signal detection, scoring, and CTD pipelines. IBM watsonx.ai integration with deterministic fallback ensures reproducible outputs in any environment. IBM Bob MCP server orchestrates 6 registered tools for seamless, intelligent workflow automation.



Thank You

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Team Lead: Vansh Vyas

